
PREDICTION OF ADVERSE DRUG REACTIONS - USING ADVANCED STATISTICAL METHODS

A PREPRINT

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Abstract

How unfortunate is that when you intake drug to cure a specific issue but end up having other adverse side effects? Adverse Drug Reactions (ADRs) represent a major challenge in pharmacovigilance and drug safety, contributing significantly to patient morbidity, mortality, and healthcare costs. So in this paper, we have evaluated and developed statistical methods and models in predicting ADRs using two complementary sources of drug information, drug gene interaction profiles and chemical fingerprints. The major challenge in working with ADR dataset is that, there is a huge imbalance in the available dataset, so the predictive model should understand the imbalance and try to figure out the pattern. The models evaluated include a Naïve Baseline, Kernel Regression (KR), Support Vector Machines (SVM) with linear and RBF kernels, and a novel Weighted Non-negative Matrix Factorisation (NMF) framework combined with Kernel Regression. Performance is assessed across 5-fold cross-validation using the Area Under the ROC Curve (AUROC) and Area Under the Precision-Recall Curve (AUPR). Our results demonstrate that the Weighted NMF with Kernel Regression achieves the highest AUPR (0.5007 ± 0.0021), making it the most effective method for handling the imbalanced nature of ADR datasets, while VKR NMF achieves the highest AUROC (0.9183 ± 0.0025). We further develop an interactive web application enabling real time single drug ADR prediction.

1 Introduction

Adverse Drug Reactions (ADRs) are unintended, harmful responses to medications administered at normal therapeutic doses. According to the WHO, ADRs are among the leading causes of hospitalisation and death worldwide, accounting for approximately 5–7% of all hospital admissions in developed countries. Beyond the human cost, ADRs impose enormous financial burdens on healthcare systems, with estimates suggesting costs in the hundreds of billions of dollars annually.

Traditional approaches to ADR detection rely primarily on post market pharmacovigilance, monitoring drug safety signals after a drug has already entered clinical use. While these methods are effective, yet someone has to experience an adverse reaction to flag the behavior. Hence, computational prediction of ADRs, offers the potential to anticipate safety issues before clinical exposure, informing drug development decisions and personalising treatment.

Drugs can be characterised by their interaction profiles with genes and by their chemical structure, both of which are expected to carry information about their biological activity and, by extension, their side effect profiles. However, the construction of such models faces two fundamental statistical challenges:

1. **Sparsity:** The drug side effect matrix is extremely sparse. Most drug side effect pairs are unobserved, not because the side effect is absent, but because it has not been reported or studied.
2. **Class Imbalance:** The number of known positive drug side effect associations is vastly smaller than the number of negatives, making standard classifiers biased toward predicting the majority (negative) class.

Addressing these challenges requires careful model selection and evaluation. Standard accuracy metrics such as AUROC can be misleading in highly imbalanced settings. Hence, AUPR is often a more informative metric when positives are rare.

In this paper, we present a comprehensive study of ADR prediction using drug gene interaction (DGI) profiles and chemical fingerprints. We evaluated a range of models from a simple Naïve Baseline to Kernel Regression, Support Vector Machines, and a novel Weighted Non-negative Matrix Factorisation (NMF) framework.

2 Data

2.1 Dataset

The primary data sources were integrated from DGI database,

- **Drug Gene Interaction (DGI) data:** We retrieved drug gene interaction (DGI) pairs from DGI 4.0 database. We mapped the drug names from SIDER 4.1 to DGIdb 4.0 and found 973 drugs in common.
- **Chemical Fingerprints:** We retrieved the drug chemical structure fingerprints (Chem) by mapping the CIDs of 1430 drugs from SIDER 4.1 to PubChem. PubChem fingerprints are binary vectors that can represent the chemical structures of drugs. We used the ChemmineR and ChemmineOB packages in Python to generate the fingerprints.
- **Side Effect data:** We used the SIDER 4.1 database. SIDER 4.1 is an ADR dataset containing drug ADR data, which provides labels for our ADR predictions. We searched features from DGIdb and PubChem for these drugs and found 973 common drugs.

2.2 Preprocessing

All three datasets were binarised (binary matrices) prior to modelling. The DGI and Chemical fingerprints data had all the drugs in the first column. Genes and generated fingerprints were presented in each columns in the respective datasets.

The final dataset comprised:

- **973 drugs** in total
- **3,647 unique genes** in the drug-gene interaction matrix
- **1,024 chemical fingerprint features** per drug

Table 1 summarises the dataset dimensions.

Table 1: Summary of dataset dimensions.

Feature Type	Dimensionality
Drug-Gene Interaction	$973 \times 3,647$
Chemical Fingerprints	$973 \times 1,024$
Side Effect Matrix	973×5779

2.3 Class Imbalance in the dataset

The drug side effect matrix is highly sparse, with a positive rate well below 10% for most side effects. This severe class imbalance motivates the use of AUPR as a primary evaluation metric alongside AUROC, since AUROC can appear inflated in imbalanced settings. In the entire dataset, about 97% of the data represents the negative class and remaining as positive class

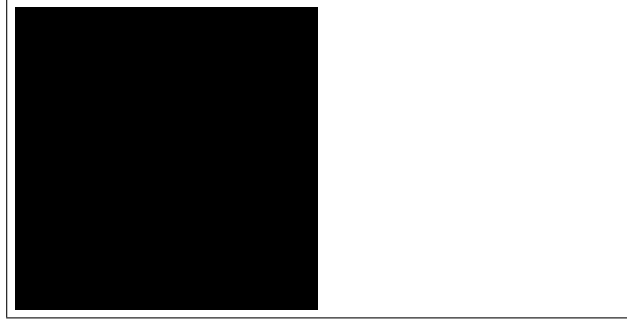


Figure 1: Sample figure caption.

Table 2: Sample table title

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	~ 100
Axon	Output terminal	~ 10
Soma	Cell body	up to 10^6

3 Examples of citations, figures, tables, references

You can insert references. Here is some text (Kour and Saabne 2014b, 2014a) and see Hadash et al. (2018).

The documentation for `natbib` may be found at

You can use custom blocks with LaTeX support from `rmarkdown` to create environment.

<http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf%7D>

Of note is the command `\citet`, which produces citations appropriate for use in inline text.

You can insert LaTeX environment directly too.

```
\citet{hasselmo} investigated\dots
```

produces

Hasselmo, et al. (1995) investigated...

<https://www.ctan.org/pkg/booktabs>

3.1 Figures

You can insert figure using LaTeX directly.

See Figure 1. Here is how you add footnotes. [[^]Sample of the first footnote.]

But you can also do that using R.

You can use `bookdown` to allow references for Tables and Figures.

3.2 Tables

Below we can see how to use tables.

See awesome Table~2 which is written directly in LaTeX in source Rmd file.

You can also use R code for that.

```
knitr::kable(head(mtcars), caption = "Head of mtcars table")
```

Table 3: Head of mtcars table

	mpg	cyl	displacement	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

3.3 Lists

- Item 1
- Item 2
- Item 3

Hadash, Guy, Einat Kermany, Boaz Carmeli, Ofer Lavi, George Kour, and Alon Jacovi. 2018. “Estimate and Replace: A Novel Approach to Integrating Deep Neural Networks with Existing Applications.” *arXiv Preprint arXiv:1804.09028*.

Kour, George, and Raid Saabne. 2014a. “Fast Classification of Handwritten on-Line Arabic Characters.” *Soft Computing and Pattern Recognition (SoCPaR), 2014 6th International Conference of*, 312–18.

Kour, George, and Raid Saabne. 2014b. “Real-Time Segmentation of on-Line Handwritten Arabic Script.” *Frontiers in Handwriting Recognition (ICFHR), 2014 14th International Conference on*, 417–22.